Linear Equations & Lines

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Review: The real number system**

Consider the system of linear inequalities represented by the shaded region in the graph below. The boundary lines are given by $y = -2x + 10$ and $y = x + 1$. The point $(k, 4)$ lies on the boundary line $y = -2x + 10$ and is within the feasible region. Determine the value of k.

The point $(k, 4)$ lies on the line $y = -2x + 10$. Substituting the coordinates of the point into the equation of the line, we get $4 = -2k + 10$. Solving for k: $2k = 10 - 4$ $2k = 6$ $k = 3$. Now we must verify that the point $(3, 4)$ is within the feasible region defined by the system of inequalities. The inequalities corresponding to the boundary lines $y = -2x + 10$ and $y = x + 1$ and the shaded region are: 1. $y \leq -2x + 10$ (since the shaded region is below or on this line, including the point (5,0) as.

2 **[Foundational]** • **Target: Review: Integer operations**

Consider the system of linear equations graphed below: $\{2x + y = 4; x - y = -1$ What is the solution (x, y) to this system?

The solution to a system of linear equations is the point where the two lines intersect. By examining the graph, we can see that the intersection point is at $(1, 2)$. We can verify this algebraically by substituting the point into both equations: For the first equation: $2(1) + 2 = 2 + 2 = 4$. This is true. For the second equation: $1 - 2 = -1$. This is true. Therefore, the solution to the system is $(1, 2)$.

3 **[Foundational]** • **Target: Review: Operations with rational numbers**

Consider the system of linear equations: $\{2x + y = 7; x - y = 2$ What is the value of x?

We are given the system of linear equations: 1) $2x + y = 7$ 2) $x - y = 2$ To find the value of x, we can add the two equations together to eliminate y: $(2x + y) + (x - y) = 7 + 2$ $2x + x + y - y = 9$ $3x = 9$ Now, divide by 3 to solve for x: $x = \frac{9}{3}$ $x = 3$ To verify, we can substitute $x = 3$ into the second equation to find y: $3 - y = 2$ $-y = 2 - 3$ $-y = -1$ $y = 1$ Now substitute $x = 3$ and $y = 1$ into the first equation: $2(3) + 1 = 6 + 1 = 7$, which is correct. Therefore, the value of x.

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4 [Foundational] • Target: Review: Exponents and radicals

Consider the system of linear equations: $\{2x + y = 5; x - y = 1$ What is the solution (x, y) for this system?

We can solve this system using the elimination method. Add the two equations together: $(2x + y) + (x - y) = 5 + 1$ $3x = 6$ Divide by 3 to find x: $x = \frac{6}{3}$ $x = 2$ Now substitute the value of x back into either of the original equations to solve for y. Using the first equation: $2(2) + y = 5$ $4 + y = 5$ Subtract 4 from both sides: $y = 5 - 4$ $y = 1$ Therefore, the solution is $(2, 1)$.

5 [Foundational] • Target: Review: Laws of exponents

Let the rational function be defined as $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$. If the graph of $y = f(x)$ has a removable discontinuity (a hole) at $x = 2$, determine the value of the parameter a.

For the rational function $f(x) = \frac{x^2 - ax + 6}{x^2 - 5x + 6}$ to have a removable discontinuity at $x = 2$, the factor $(x - 2)$ must be present in both the numerator and the denominator, and it must cancel out. First, factor the denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$. For $(x - 2)$ to be a factor of the numerator $x^2 - ax + 6$, substituting $x = 2$ into the numerator must result in zero: $(2)^2 - a(2) + 6 = 0$. This simplifies to $4 - 2a + 6 = 0$, which gives $10 - 2a = 0$. Solving for a, we get $2a = 10$.

6 [Foundational] • Target: Units resulting from a formula or graph

Consider the function $f(x) = x^2 - 4x + 7$. The graph of $y = f(x)$ is subjected to a sequence of transformations. First, it is translated by a units horizontally and b units vertically. Let this transformed function be $g(x)$. Subsequently, the graph of $y = g(x)$ is reflected in the x-axis to produce the graph of $y = h(x)$. If the vertex of the graph of $y = h(x)$ is at $(-2, -3)$, determine the value of $a^2 + b^2$.

The original function is $f(x) = x^2 - 4x + 7$. We can complete the square to find the vertex: $f(x) = (x^2 - 4x + 4) + 3 = (x - 2)^2 + 3$. The vertex of $y = f(x)$ is at $(2, 3)$. When $y = f(x)$ is translated by a units horizontally and b units vertically, the new function $g(x)$ is given by $g(x) = f(x - a) + b$. The vertex of $g(x)$ will be at $(2 + a, 3 + b)$. Next, the graph of $y = g(x)$ is reflected in the x-axis to produce $y = h(x)$. This means $h(x) = -g(x)$. The vertex of $h(x)$ will be at $(2 + a, -(3 + b))$. We are g.

7 [Foundational] • Target: Review: The real number system

The graph of the linear equation $y = mx + c$ is shown below. The line passes through the point $(-3, -11)$ and intersects the y-axis at $(0, k)$. If the slope of the line is -3, determine the value of k.

The equation of a line is given by $y = mx + c$, where m is the slope and c is the y-intercept. We are given that the slope $m = -3$. Therefore, the equation of the line is $y = -3x + c$. The line passes through the point $(-3, -11)$. We can substitute these coordinates into the equation to find c: $-11 = -3(-3) + c$ $-11 = 9 + c$ $c = -11 - 9$ $c = -20$ The y-intercept is the point where the line crosses the y-axis, which is $(0, k)$. For the equation $y = -3x + c$, when $x = 0$, $y = c$. Thus, $k = c$. Since.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Review: Integer operations**

What is the solution (x, y) to the system of equations below? $x + y, = 8; 2x - y, = 7$

To solve the system of linear equations, we can use the elimination method by adding the two equations together: $(x + y) + (2x - y) = 8 + 7$ Combine like terms: $3x = 15 \Rightarrow x = 5$ Substitute $x = 5$ into the first equation to find y: $5 + y = 8 \Rightarrow y = 3$ Thus, the solution to the system is $(5, 3)$. - $(3, 5)$ results from reversing the x- and y-coordinates. - $(4, 1)$ satisfies the second equation $(2(4) - 1 = 7)$ but fails the first ($4 + 1 = 5eq8$). - $(6, 2)$ satisfies the first equat.

9 **[Intermediate]** • **Target: Review: Operations with rational numbers**

Consider the polynomial $P(x) = x^3 - 13x + 12$. The graph of $y = P(x)$ is shown below. The polynomial can be factored into the form $P(x) = (x - a)(x - b)(x - c)$ where $a < b < c$. Determine the value of $c - a$.

The problem requires factoring a cubic polynomial and interpreting the roots from a graph to solve for a difference between roots. First, we identify possible rational roots of $P(x) = x^3 - 13x + 12$ using the Rational Root Theorem. Possible rational roots are divisors of the constant term (12) divided by divisors of the leading coefficient (1), which are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$. From the graph, we can visually estimate the x-intercepts (roots) of the polynomial. The graph shows roots at.

10 **[Intermediate]** • **Target: Review: Exponents and radicals**

The graph of the function $f(x) = ax^2 + b$ is shown below. The graph passes through the points $(-2, 13)$ and $(3, 28)$. Determine the value of b.

The function is given by $f(x) = ax^2 + b$. We are given two points that lie on the graph of this function: $(-2, 13)$ and $(3, 28)$. Substitute the first point into the equation: $13 = a(-2)^2 + b$, which simplifies to $13 = 4a + b$. (Equation 1) Substitute the second point into the equation: $28 = a(3)^2 + b$, which simplifies to $28 = 9a + b$. (Equation 2) Now we have a system of two linear equations with two variables, a and b: 1) $4a + b = 13$ 2) $9a + b = 28$ To find the value of b, we can eliminate a.

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11 **[Intermediate]** • **Target: Review: Laws of exponents**

The bar graph shows the frequency distribution of scores obtained by 20 students on a 5-point quiz, where no student originally scored 5. A group of k additional students take the quiz later, and each of these students earns a score of 5. If the mean score of the combined group of $20 + k$ students is equal to 3, what is the value of k?

Step 1: Calculate the total sum of the scores for the original 20 students from the bar graph: $\text{Sum}_{\text{original}} = (1 \times 4) + (2 \times 7) + (3 \times 6) + (4 \times 3) = 4 + 14 + 18 + 12 = 48$ Step 2: Add the k additional students, each scoring 5 points. The new sum of scores is $48 + 5k$, and the new total number of students is $20 + k$. Step 3: Set up the equation for the new mean: $\frac{48 + 5k}{20 + k} = 3$ Step 4: Solve for k: $48 + 5k = 3(20 + k)$ $48 + 5k = 60 + 3k$.

12 **[Intermediate]** • **Target: Units resulting from a formula or graph**

The graph of the function $f(x) = x^2$ is shown below. It is transformed to the graph of $g(x) = (x - 3)^2 + 2$. Which transformation maps $f(x)$ to $g(x)$?

The function $g(x) = (x - 3)^2 + 2$ is a transformation of the base function $f(x) = x^2$. In the general form of a quadratic transformation, $h(x) = a(x - h)^2 + k$, a value of h represents a horizontal translation, and a value of k represents a vertical translation. If h is positive, the translation is to the right. If h is negative, the translation is to the left. If k is positive, the translation is upwards. If k is negative, the translation is downwards. In $g(x) = (x - 3)^2 + 2$, we have $h = 3$ and.

13 **[Intermediate]** • **Target: Review: The real number system**

The graph of the function $f(x) = a \cdot b^x + c$ is shown below. The graph passes through the points $(0, 1)$ and $(1, 7)$, and has a horizontal asymptote at $y = -5$. Determine the values of a and b.

The function is given by $f(x) = a \cdot b^x + c$. The horizontal asymptote is at $y = -5$, which means $c = -5$. So, the function is $f(x) = a \cdot b^x - 5$. We are given that the graph passes through the point $(0, 1)$. Substituting these coordinates into the function: $1 = a \cdot b^0 - 5$ $1 = a \cdot 1 - 5$ $1 = a - 5$ $a = 6$ Now we know $a = 6$, so the function is $f(x) = 6 \cdot b^x - 5$. We are also given that the graph passes through the point $(1, 7)$. Substituting these coordinates into the function.

14 **[Intermediate]** • **Target: Review: Integer operations**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(1, 5)$ and the parabola passes through the point $(3, -3)$. Determine the value of a.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. \\ Given the vertex is $(1, 5)$, the function can be written as $f(x) = a(x - 1)^2 + 5$. \\ The graph also passes through the point $(3, -3)$. We can substitute these coordinates into the equation to solve for a: \\ $-3 = a(3 - 1)^2 + 5$ \\ $-3 = a(2)^2 + 5$ \\ $-3 = 4a + 5$ \\ Subtract 5 from both sides: \\ $-3 - 5 = 4a$ \\ $-8 = 4a$ \\ Divide by 4: \\ $a = \frac{-8}{4}$ \\ $a = -2$. \\

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 4.01 Functions & Transformations

The graph of the quadratic function $f(x) = ax^2 + bx - 5$ is shown below. The function has roots at $x = -5$ and $x = 1$. Find the value of a.

Since the roots of the quadratic function $f(x) = ax^2 + bx - 5$ are $x = -5$ and $x = 1$, we can write the factored form of the quadratic as $f(x) = a(x - (-5))(x - 1) = a(x + 5)(x - 1)$. Expanding this, we get $f(x) = a(x^2 - x + 5x - 5) = a(x^2 + 4x - 5) = ax^2 + 4ax - 5a$. We are given that the constant term is -5. Therefore, $-5a = -5$, which implies $a = 1$. The quadratic function is $f(x) = 1x^2 + 4(1)x - 5 = x^2 + 4x - 5$. This is consistent with the given information.

16 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The bar chart below shows the number of students who participated in different sports at a school. What is the total number of students who participated in basketball and volleyball?

The bar chart shows that 22 students participated in basketball and 18 students participated in volleyball. To find the total number of students who participated in these two sports, we add the number of participants for each sport: 22 (basketball) + 18 (volleyball) = 40 students.

17 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 6.01 Systems & Inequalities

The following bar chart shows the number of students who achieved different scores on a recent quiz. What is the mode of the quiz scores?

The mode of a dataset is the value that appears most frequently. In the given bar chart, the score of 80 has the highest frequency, with 25 students achieving this score. Therefore, the mode of the quiz scores is 80.

18 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 7.01 Sequences & Series

The bar chart below shows the number of students in four different clubs at a school. What is the total number of students in the Science Club and the Chess Club combined?

From the bar chart, the number of students in the Science Club is 18 and the number of students in the Chess Club is 12. The total number of students in both clubs is the sum of these two values: $18 + 12 = 30$.

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19 [Challenging] • Target: Review: The real number system • Cross-Lesson: 8.01 Factoring Polynomials

Let S be the set of all distinct real solutions to the equation $(x^2 - x + 1)^4 - 10x^2(x^2 - x + 1)^2 + 9x^4 = 0$. What is the sum of the squares of the elements of S?

To solve the equation without expanding the degree-8 polynomial, let $u = x^2 - x + 1$. Substituting u into the equation yields a quadratic form in u^2: $u^4 - 10x^2u^2 + 9x^4 = 0$. Factoring this as a quadratic in u^2: $(u^2 - x^2)(u^2 - 9x^2) = 0$. Applying the difference of squares factorization to each factor gives: $(u - x)(u + x)(u - 3x)(u + 3x) = 0$. Now substitute $u = x^2 - x + 1$ back into each linear factor in u: 1. $u - x = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0$, giving the re.

20 [Challenging] • Target: Review: Integer operations • Cross-Lesson: 9.01 Radicals & Exponents

The rational function $R(x) = \frac{x^2 - 7x + k}{x - 3}$ has a removable discontinuity (a hole) at $x = 3$. For what value of k does this occur?

A rational function of the form $R(x) = \frac{P(x)}{Q(x)}$ has a removable discontinuity (a hole) at $x = a$ if $(x - a)$ is a factor of both the numerator $P(x)$ and the denominator $Q(x)$. In this case, the denominator is $Q(x) = x - 3$, so the discontinuity occurs at $x = 3$. For this discontinuity to be removable, $(x - 3)$ must also be a factor of the numerator, $P(x) = x^2 - 7x + k$. This means that when $x = 3$, $P(3)$ must equal zero. We can substitute $x = 3$ into the numerator and set the expression equal t.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

22 [Foundational] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(5)$.

a. 13 ✓

b. 15

c. 10

d. 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

24 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

25 [Foundational] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

27 [Foundational] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

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28 [Foundational] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Review: The real number system

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

31 [Intermediate] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(4)$.

a. 11 ✓

b. 13

c. 8

d. 15

Substitute $x = 4$: $f(4) = 2(4) + 3 = 11$.

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32 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

33 [Intermediate] • Target: Review: Integer operations

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

34 [Intermediate] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

35 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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36 [Intermediate] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

37 [Intermediate] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(4)$.

a. 21 ✓

b. 25

c. 16

d. 25

Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

39 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

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40 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

41 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

42 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

43 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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44 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

45 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

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